PROBABILITY

From Basic Probability to Probability Distributions

DATA ANALYTICS BY DURGA

Basic probability • Conditional probability • Counting • Tree & Venn diagrams • Binomial • Poisson • Bayes' theorem • Excel

Learning Objectives

Understand probability terminology and the probability scale 0 to 1.

Define experiment, chance experiment, outcome, sample space and event.

Calculate probabilities using Addition and Multiplication Rules.

Determine whether events are mutually exclusive or independent.

Calculate and interpret conditional probabilities.

Construct and interpret contingency tables, tree diagrams and Venn diagrams.

Use counting rules, permutations and combinations.

Compute probabilities from Binomial and Poisson distributions.

Apply Bayes' theorem to reverse conditional probabilities.

xi	fi	xifi
0	35	0
1	25	25
2	20	40
3	10	30
4	5	20
5	5	25
	100	140
	Mean	1.4

Probability: Meaning and Scale

Probability

A numerical measure of the chance, likelihood, or possibility that an event will occur.

Scale

0 = impossible • 1 = certain • 0.5 = equally likely to occur or not occur.

$$0 \leq P(A) \leq 1$$

For equally likely outcomes: favorable outcomes ÷ total possible outcomes.

Experiment, Outcome and Sample Space

Experiment: a planned operation under controlled conditions.

Chance experiment: the result is not predetermined.

Outcome: a result of an experiment.

Sample space S: the set of all possible outcomes.

A sample space may be represented by a list, tree diagram or Venn diagram.

Example

One fair coin: S = {H, T}.

Events and Complements

Event: any combination of outcomes; A, B, C, … represent events.

Simple event: one characteristic.

Joint event: two or more characteristics.

Complement A′: all outcomes not in A.

Complement rule: P(A)+P(A′)=1.

$$P(A') = 1 - P(A)$$

Simple Probability Example

Two fair coins are tossed.

S = {HH, HT, TH, TT}.

A = exactly one head = {HT, TH}.

There are 2 favorable outcomes out of 4 equally likely outcomes.

$$P(A)=2/4=0.50$$

EVENT RELATIONSHIPS

AND • OR • COMPLEMENT • EXCLUSIVITY • INDEPENDENCE

A OR B vs A AND B

A OR B

In A, in B, or in both. $A \cup B$

A AND B

In both A and B at the same time. $A \cap B$

Example: A={1,2,3,4,5}, B={4,5,6,7,8}.

A OR B={1,2,3,4,5,6,7,8}.

A AND B={4,5}.

Mutually Exclusive Events

Two events are mutually exclusive if they cannot occur simultaneously.

They have no outcomes in common.

Therefore $P(A \cap B)=0$.

Example: choosing one day from a year — January and February are mutually exclusive.

Mutually exclusive $\Rightarrow P(A \cap B)=0$

Collectively Exhaustive Events

A set of events is collectively exhaustive when it covers the entire sample space.

At least one event in the set must occur.

If the events are also mutually exclusive, exactly one occurs.

P(A)+P(B)+P(C)=1 (if A,B,C are mutually exclusive & exhaustive)

Independence vs Dependence

Independent events: one event occurring does not affect the probability of the other.

Checks: $P(A|B)=P(A)$, $P(B|A)=P(B)$, or $P(A \cap B)=P(A)P(B)$.

Dependent events: the occurrence of one event changes the probability of the other.

Sampling clue

With replacement → typically independent. Without replacement → typically dependent.

PROBABILITY RULES

ADDITION • MULTIPLICATION • CONDITIONAL

General Addition Rule

$$P(A \text{ OR } B) = P(A) + P(B) - P(A \text{ AND } B)$$

Subtract the overlap because $A \cap B$ would otherwise be counted twice.

If A and B are mutually exclusive, $P(A \cap B) = 0$ and the rule becomes $P(A \text{ OR } B) = P(A) + P(B)$.

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Multiplication Rule

$$P(A \text{ AND } B) = P(A) \times P(B|A)$$

Use multiplication when the question contains AND.

For independent events, $P(B|A)=P(B)$.

Therefore independent: $P(A \cap B)=P(A)P(B)$.

Conditional Probability

$$P(A|B)=P(A \text{ AND } B)/P(B)$$

P(A|B) is the probability of A given B has already occurred.

The condition reduces the sample space to B.

Use contingency tables or Venn diagrams to visualize the reduced sample space.

Conditional Probability: Car Example

70% of used cars have AC; 40% have GPS; 20% have both.

Find P(GPS|AC).

Given AC, the relevant sample space is the 70% of cars with AC.

P(GPS|AC)=0.20/0.70=0.2857 ≈ 28.57%.

P(GPS|AC)=0.20/0.70=0.2857

VISUALIZING PROBABILITY

CONTINGENCY TABLES • TREE DIAGRAMS • VENN DIAGRAMS

Contingency Tables

A contingency table organizes outcomes using two variables.

Row/column totals support marginal probabilities.

Interior cells support joint probabilities.

Conditional probabilities use a reduced row or column as the denominator.

	Violation	No violation	Total
Uses phone	25	280	305
Does not use	45	405	450
Total	70	685	755

Reading a Contingency Table

P(cell-phone user)=305/755.

P(no violation)=685/755.

P(no violation AND cell-phone user)=280/755.

P(cell-phone user | violation)=25/70.

Key idea

For conditional probability, the denominator is the size of the reduced sample space.

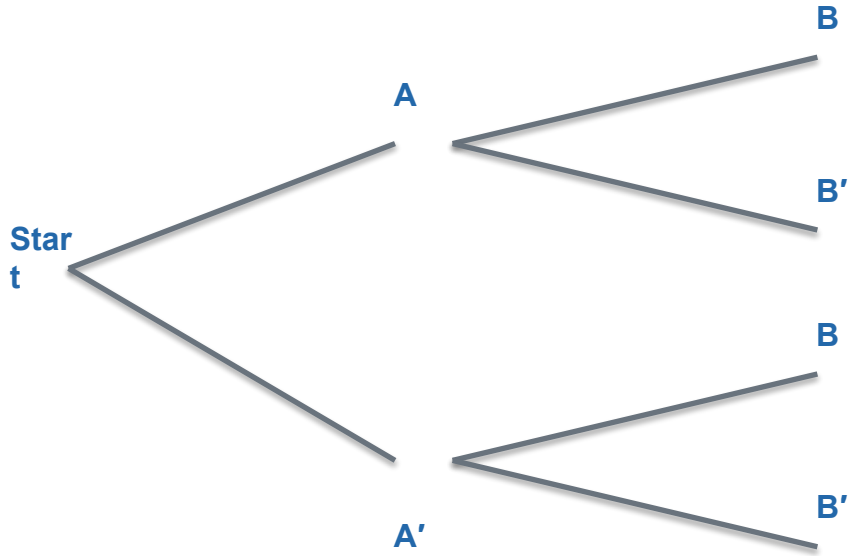
Tree Diagrams

A tree diagram is a graph used to determine experiment outcomes.

Branches may show frequencies or probabilities.

Multiply along a path for AND.

Add alternative paths for OR.



Venn Diagrams

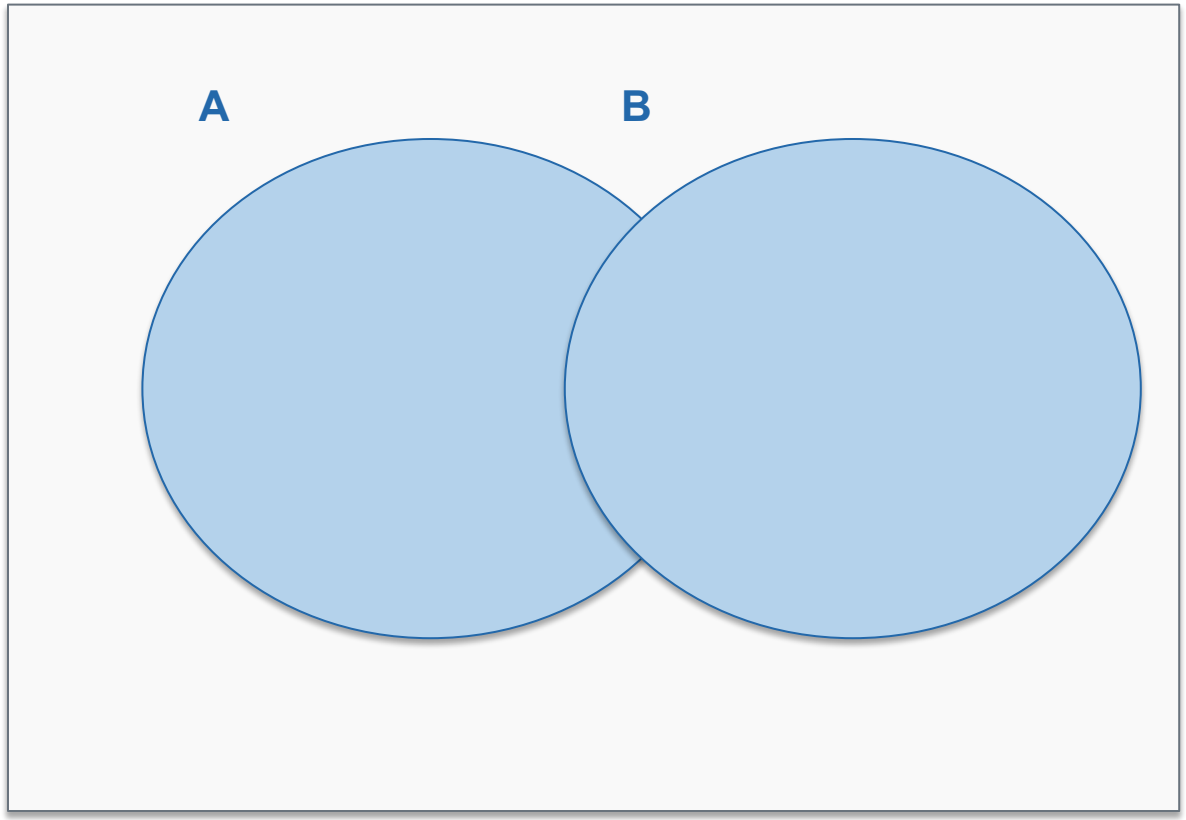
Rectangle = sample space S.

Circles = events.

Overlap = A AND B (A∩B).

Combined circle area = A OR B (A∪B).

Outside A = A′.



COUNTING METHODS

COUNTING RULES • FACTORIAL • PERMUTATIONS • COMBINATIONS

Counting Rule 1: Repeated Trials

Number of possible outcomes = k^n

If k possible outcomes can occur on each of n trials, there are k^n possible outcomes.

Coin tossed 5 times → $2^5=32$.

Die rolled twice → $6^2=36$.

Counting Rule 2: Different Choices

Total outcomes=$(k_1)(k_2)\ldots(k_{\square})$

Example: 5 appetizers × 10 entrées × 3 beverages × 6 desserts.

Total dinner choices = 5×10×3×6 = 900.

Counting Rule 3: Factorial

$$n! = n(n-1)\ldots(2)(1), \quad 0! = 1$$

Counts arrangements of all n distinct items.

Six books on a shelf → 6!=720 arrangements.

Order matters.

Permutations: Order Matters

$$^{n}P_{x} = n!/(n-x)!$$

Use when x objects are selected from n objects and order matters.

Example: arrange 4 books from 6 → $^{6}P_{4}=360$.

Typical cues: ranking, seating positions, codes, finishing order.

Combinations: Order Does Not Matter

$$^{n}C_{x} = n!/[x!(n-x)!]$$

Use when x objects are selected from n objects and order does not matter.

Example: choose 3 flavors from 31 → $^{31}C_{3}$=4,495.

Typical cues: committees, teams, selections.

Permutation or Combination?

Question	Order?	Method
Choose 3 students for a committee	No	Combination
Rank top 3 finishers	Yes	Permutation
Select 4 products for a bundle	No	Combination
Create a 4-digit arrangement	Yes	Permutation

Quick test

Would changing the order create a different outcome?

DISCRETE PROBABILITY DISTRIBUTIONS

RANDOM VARIABLES • EXPECTED VALUE • VARIANCE • BINOMIAL • POISSON

Discrete vs Continuous Random Variables

Type	Definition	Examples
Discrete	Counting process	No. of classes; no. of heads
Continuous	Measurement	Salary; weight; waiting time

Recognition cue

Discrete = how many? Continuous = how much / how long?

Probability Distribution of a Discrete Variable

A probability distribution lists all possible numerical outcomes and their probabilities.

Probabilities are between 0 and 1.

All probabilities must sum to 1.

X	P(X)
0	0.35
1	0.25
2	0.2
3	0.1
4	0.05
5	0.05

$$\Sigma P(X)=1$$

Expected Value: Weighted Average

$$E(X)=\mu=\Sigma x_i P(X=x_i)$$

Expected value is the long-run mean of a discrete random variable.

Two fair coin tosses, X=#heads: E(X)=(0)(.25)+(1)(.50)+(2)(.25)=1.

Variance and Standard Deviation

$$\sigma^2=\Sigma[x_i-E(X)]^2P(X=x_i) \quad \text{and} \quad \sigma=\sqrt{\sigma^2}$$

Variance measures weighted squared distance from the expected value.

Standard deviation is the square root of variance.

Network-interruption example from the source: $\mu=1.40$, $\sigma^2=2.04$, $\sigma\approx1.4283$.

Binomial Distribution

Models the number of successes in a fixed number n of independent trials.

Each trial has two mutually exclusive and exhaustive outcomes.

Probability of success π is constant.

Trials are independent.

Applications: defective/acceptable items, contract won/lost, yes/no survey response.

Recognition

Fixed n + two outcomes + constant p + independence → BINOMIAL.

Binomial Probability Formula

$$P(X=x|n,\pi)=C(n,x)\pi^{x}(1-\pi)^{n-x}$$

X = number of events of interest.

n = number of trials.

π = probability of event of interest.

1−π = probability of no event of interest.

Binomial Worked Example

Probability an invoice payment is late = 0.10.

Find probability of exactly 1 late payment in 4 invoices.

n=4, x=1, π=.10.

$$P(X=1)=C(4,1)(.10)^{1}(.90)^{3}=.2916$$

Binomial in Excel

Excel function: BINOM.DIST(x,n,p,cumulative).

For exact probability, use FALSE.

Example: x=1, n=4, p=.10.

=BINOM.DIST(1,4,0.10,FALSE)

Poisson Distribution

Counts the number of times an event occurs in a given area of opportunity.

Area can be a time interval, distance, area, or volume.

Events are independent.

Average number of events per unit is λ.

Poisson Formula

$$P(X=x|\lambda)=e^{-\lambda}\,\lambda^{x}/x!$$

X = number of events.

λ = expected number of events.

e≈2.71828.

Poisson Worked Example

Average customer arrival rate = 0.5 per minute.

Find probability of exactly 2 arrivals in a minute.

x=2, λ=.50.

$$P(X=2|\lambda=.50)=e^{-0\cdot5}(.50)^2/2!=.0758$$

Poisson in Excel

Excel function: POISSON.DIST(x,mean,cumulative).

For exact probability, use FALSE.

Example: x=2, mean=.50.

=POISSON.DIST(2,0.50,FALSE)

BAYES' THEOREM

REVERSE CONDITIONAL PROBABILITY

Bayes' Theorem: Core Idea

Bayes' theorem updates a probability when new evidence is observed.

It reverses a conditional probability.

Applications include diagnosis, fraud detection, classification and risk analysis.

$$P(B|A)=P(B)P(A|B)/P(A)$$

Bayes Worked Example: Diagnostic Test

Disease prevalence P(C)=0.03.

Positive test when disease is present P(YES|C)=0.99.

False positive rate P(YES|NC)=0.02.

Find P(C|YES).

P(C|YES)=(.03×.99)/[(.99×.03)+(.02×.97)]=.6049

APPLICATION & EXCEL

EMPIRICAL PROBABILITY • ETHICS • MODEL SELECTION • PRACTICE

Three Approaches to Assessing Probability

Approach	Basis	Typical use
A priori	Prior knowledge / equally likely outcomes	Dice, cards
Empirical	Observed relative frequency	Historical data
Subjective	Experience, opinion & analysis	Forecasting / judgment

Data analytics

Historical observations can be converted into empirical probabilities for operational decisions.

Ethics: Communicating Probability

Probability statements can be misunderstood.

Advertising can intentionally or unintentionally present probabilities in misleading ways.

State the denominator, population, time period and condition.

Distinguish P(positive test) from P(disease | positive test).

Principle

A mathematically correct probability can still support a poor decision if context is hidden.

Integrated Model Selection

Scenario	Model / method	Reason
10 coin tosses; count heads	Binomial	Fixed n, two outcomes, independent
Customers per minute	Poisson	Count events in interval
Choose 4 students from 20	Combination	Order not important
Rank top 3 candidates	Permutation	Order matters
Update disease probability after test	Bayes	Reverse conditional probability

Tip

Before calculating, identify the structure of the problem.

Excel Probability Toolkit

Task	Excel formula
Basic probability	=favorable/total
Conditional probability	=joint/marginal
Binomial exact	=BINOM.DIST(x,n,p,FALSE)
Binomial cumulative	=BINOM.DIST(x,n,p,TRUE)
Poisson exact	=POISSON.DIST(x,mean,FALSE)
Factorial	=FACT(n)
Combination	=COMBIN(n,x)
Permutation	=PERMUT(n,x)

Mini Excel Lab: Binomial Distribution

Enter X values 0–10.

Enter n and p in separate cells.

Use BINOM.DIST(X,n,p,FALSE) for each X.

Check that probabilities sum to approximately 1.

Create a column chart of X vs P(X).

Interpret the most likely outcome and compare it with E(X)=np.

Example: n=10, p=.30 → E(X)=np=3

Mini Excel Lab: Poisson Distribution

Set λ=2 events per hour.

Enter X values 0–8.

Use POISSON.DIST(X,2,FALSE).

Check the probability total.

Plot X against P(X).

Interpret the distribution for operational demand.

Example: λ=2 events/hour → E(X)=λ=2

Practice: Cellphone Selection

A box has 9 cellphones: 2 gold and 7 silver.

Select 2 without replacement.

Find probability of 1 gold AND 1 silver.

Because the first phone is not replaced, the events are dependent.

$$P(GS \text{ or } SG)=(2/9)(7/8)+(7/9)(2/8)=7/18\approx0.3889$$

Probability Formula Sheet

Concept	Formula
Complement	$P(A') = 1 - P(A)$
Addition	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$
Mutually exclusive	$P(A \cap B) = 0$
Conditional	$P(A\|B) = P(A \cap B)/P(B)$
Multiplication	$P(A \cap B) = P(A)P(B\|A)$
Independent	$P(A \cap B) = P(A)P(B)$
Binomial	$C(n,x)p^x(1-p)^{n-x}$
Poisson	$e^{-}\lambda\ \lambda^x/x!$
Bayes	$P(B\|A) = P(B)P(A\|B)/P(A)$

Probability Decision Framework

1. Define the experiment and event.
2. Identify the sample space.
3. Check whether outcomes are equally likely.
4. Look for AND / OR / GIVEN.
5. Check mutual exclusivity and independence.
6. If counting is needed, choose factorial / permutation / combination.
7. For fixed independent success counts, consider Binomial.
8. For event counts in an interval, consider Poisson.
9. If evidence reverses a conditional, consider Bayes.
10. Validate: probability must be between 0 and 1 and interpret it in context.

Key Takeaways

Probability quantifies uncertainty and supports data-driven decisions.

Sample spaces and events are foundational.

Addition handles OR; multiplication handles AND; conditional handles GIVEN.

Mutual exclusivity and independence are different concepts.

Contingency tables, tree diagrams and Venn diagrams make probability visual.

Counting rules support large outcome spaces.

Binomial models fixed independent trials; Poisson models event counts in an opportunity interval.

Bayes updates beliefs using evidence.

Excel can automate calculations and visualize discrete distributions.

Classroom Challenge

Challenge 1

A company receives an average of 3 complaints per day. What is the probability of exactly 5 complaints tomorrow?

Challenge 2

A product has a 5% defect probability. In 20 products, what is the probability of exactly 2 defective products?

Challenge 3

A screening test is positive 98% of the time when a condition is present, but the condition is rare. How would Bayes' theorem help interpret a positive result?

Show your work

Model selection → formula → Excel calculation → interpretation → business recommendation.

Source Basis

Week 3 Learning Material — Basic Probability: probability concepts, sample space, events, simple/joint/marginal probability, mutually exclusive and collectively exhaustive events, addition rule, conditional probability, independence, multiplication rule and ethical communication.

Week 3 Learning Material 2 — Introductory Business Statistics, Chapter 3: probability terminology, contingency tables, tree diagrams, Venn diagrams, counting rules, permutations, combinations, Binomial, Poisson and Bayes' theorem.

Week 4 Learning Material — Discrete Probability Distribution: discrete random variables, probability distributions, expected value, variance/standard deviation, counting rules, Binomial and Poisson distributions and Excel examples.

The instructional sequence and examples in this deck are adapted from the three uploaded files for a Data Analytics classroom.

Branding

Data Analytics by Durga